

NeuroForge **Nexus**

Enterprise Software Lifecycle and DevOps Management
System

TEAM 4

Introduction



What We Built

NeuroForge Nexus is an all-in-one centralized workspace designed to help software teams manage their entire development process from start to finish.



The Problem It Solves

Instead of teams juggling multiple different tools for planning tasks, testing code, and launching products, our platform brings everything together into a single, unified dashboard.



How It Helps

It creates a smooth, automated pipeline where managers can track team progress, developers can safely test their code, and operations teams can monitor the live application's health without missing a beat.

Project Overview

Secure Access Management

The platform uses a strict Role-Based Access Control (RBAC) system powered by Keycloak to ensure everyone has the right level of access to the workspace.

Administrator Role (Full Control)

Admins and Project Manager act as the controllers of the platform. They have the exclusive authority to create new projects, build teams, assign specific people to those teams, and make global edits to the system settings.

Team Member Roles (Restricted Access)

Other roles have tailored, restricted access. They cannot alter team structures or edit core project settings.

Core Platform Features

Secure Identity & Access

Powered by Keycloak for strict Role-Based Access Control (RBAC) across Admin, PM, Dev, and QA roles.

Agile Tracking & Event Streaming

Features interactive Kanban boards with a 94% burndown success rate, backed by Apache Kafka for real-time task notifications.

Automated CI/CD Pipelines

Integrates Jenkins and GitHub Actions for seamless build and deployment tracking, achieving a 97.4% success rate and rapid 4.2-minute deploys.

DevOps Observability

Utilizes Prometheus, Grafana, and the ELK Stack to continuously monitor service health, CPU, and memory usage.

Advanced Release Management

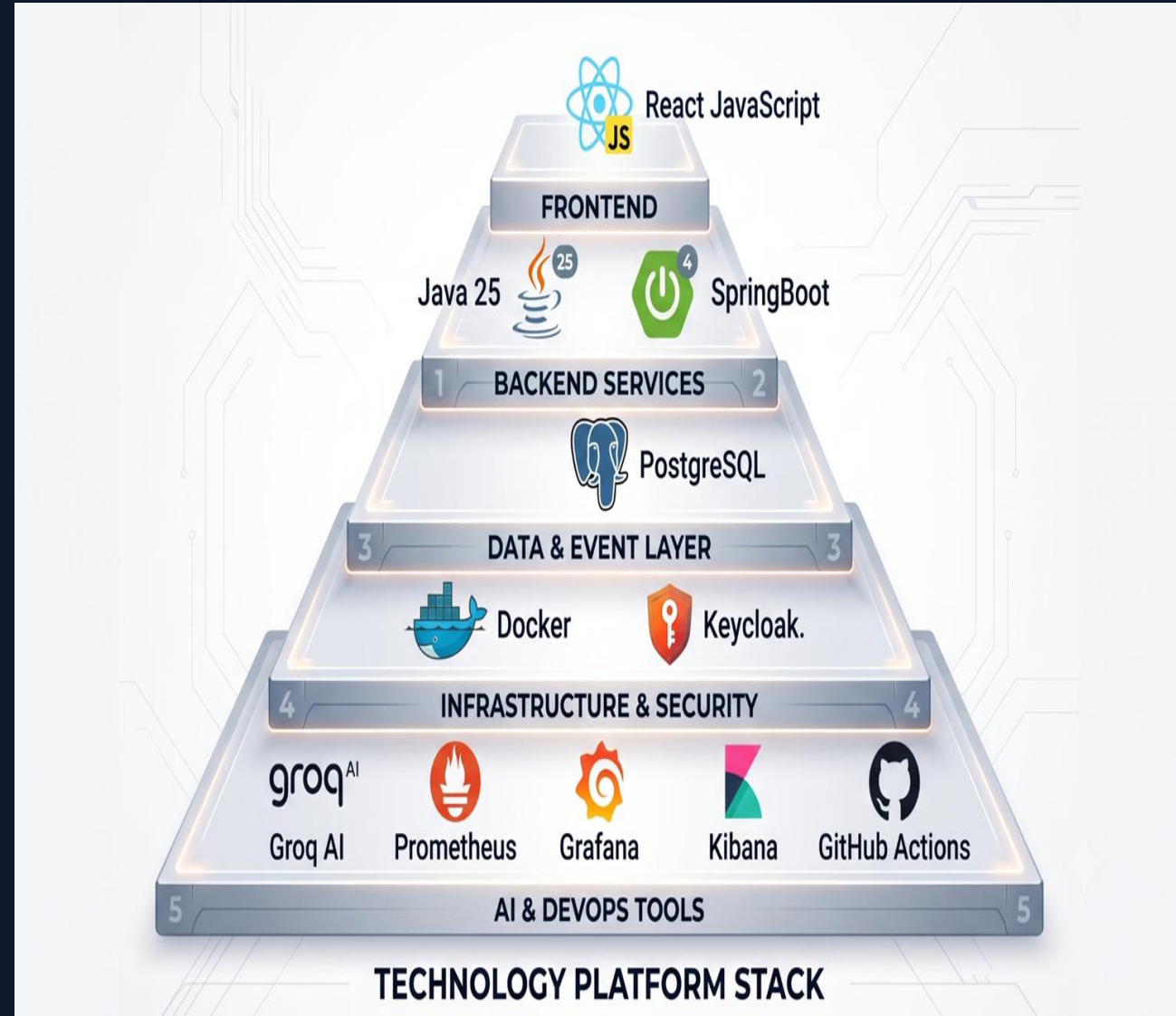
Supports Blue-Green deployments and instant rollback capabilities to maintain a lightning-fast 12-minute MTTR (Mean Time To Recovery).

AI-Powered Assistance

Features an integrated Groq AI Assistant directly within the dashboard to provide smart insights and streamline platform workflows.

Technology Stack & Architecture

Frontend	React, JavaScript
Backend Services	Java 25, Spring Boot 4
Data & Event Layer	PostgreSQL
Infrastructure & Security	Docker, Keycloak.
AI & DevOps Tools	Groq AI, Prometheus, Grafana, Kibana, and GitHub Actions



| Project Milestones

Milestone 1 (Weeks 1-2)

Project & User Management
(Implementing RBAC, Keycloak IAM, and Team Setup).

Milestone 2 (Weeks 3-4)

Sprint & Task Management
(Deploying Kanban boards, velocity tracking, and blocker management).

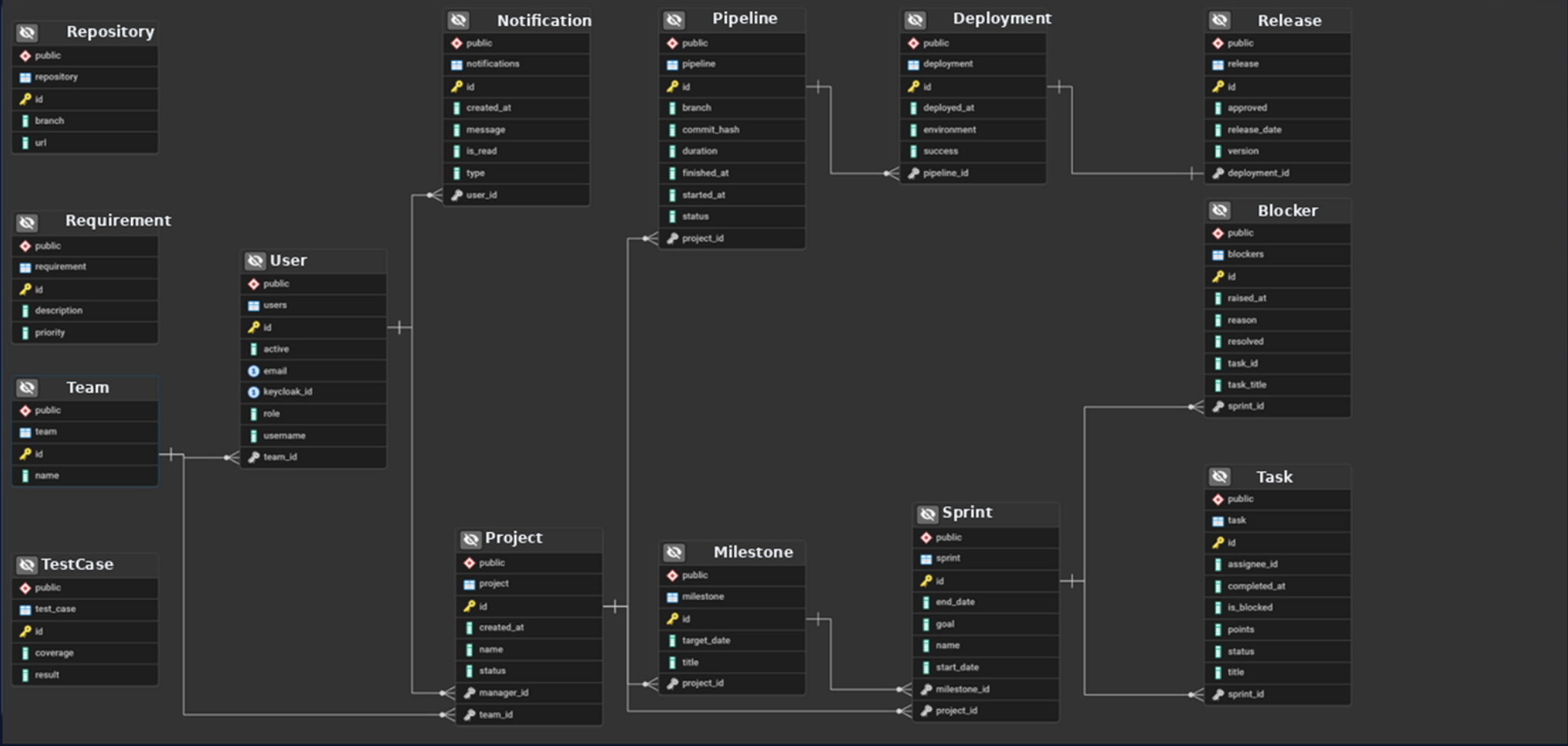
Milestone 3 (Weeks 5-6)

CI/CD Pipeline
(Integrating Jenkins, tracking builds, test results, and deployment status).

Milestone 4 (Weeks 7-8)

Release & Monitoring
(Deploying DevOps observability, alert management, and log analysis).

Class diagram



🔗 CLOUD-NATIVE SDLC PLATFORM

Ship software with **one** **connected platform**

NeuroForge Nexus brings project tracking, agile sprints and DevOps visibility into a single enterprise-grade workspace — built on Spring Boot, Kafka and Keycloak.

[Login / Register](#)

Project Tracking

Agile Sprints

Kanban & Burndown

DevOps Monitoring

Register & Login

Create your account

Join NeuroForge Nexus to start collaborating

Username *

Email *

First name *

Last name *

Select Your Role *

Select Select Your Role



Password *



At least 8 characters, with an uppercase letter, a lowercase letter, a number, and a symbol.

Confirm password *



[« Back to Login](#)

Register



Welcome back

Sign in to continue to NeuroForge Nexus

Username or email

Password



[Forgot Password?](#)

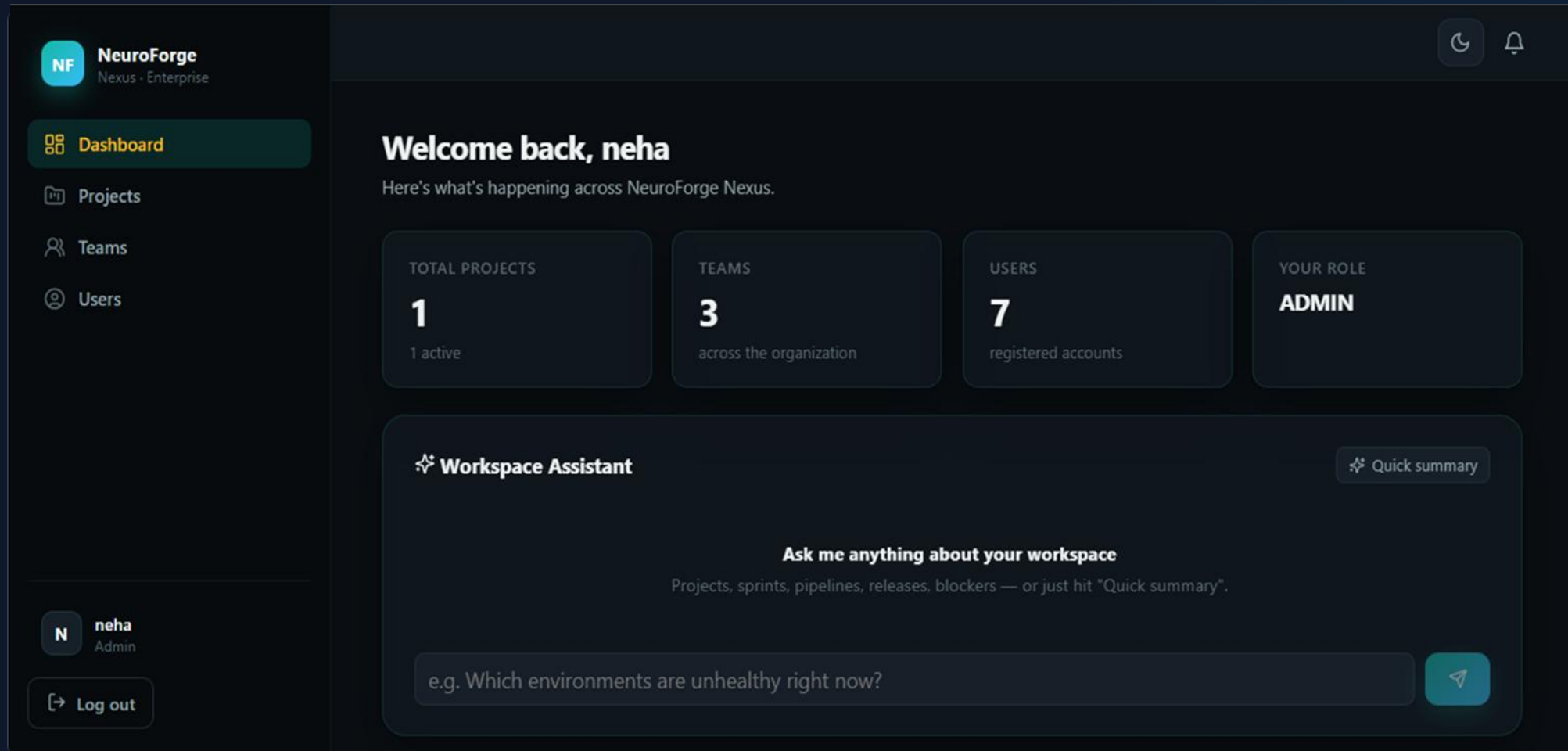
Sign In

New user? [Register](#)



Dashboard

Centralized Project Dashboard: Managing active projects, teams, status updates, and granular project settings from a unified command center.





Dashboard



Projects



Teams



Users

Recent Projects

[View all →](#)

NAME	STATUS	TEAM	MANAGER	CREATED
NeuroForge	Active	Frontend	neha	8/18/2026

Analytics

Project:

NeuroForge



Milestone:

All Milestones (Project View)



Overall Project Progress

0% Completed (0 / 0 pts)

SPRINTS TRACKED

4

TOTAL TASKS

0

TOTAL POINTS
COMMITTED

0 pts

RESOLVED / TOTAL
BLOCKERS

1 / 1

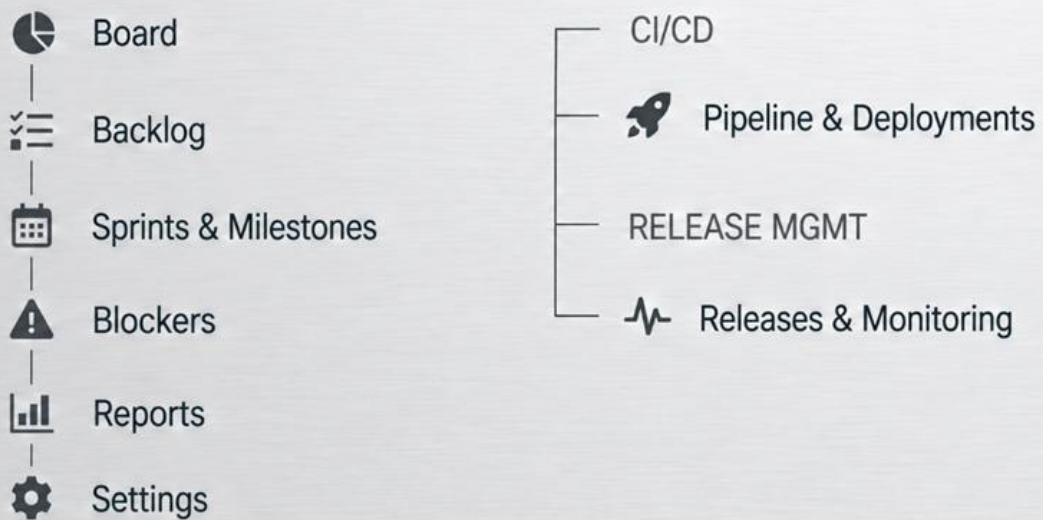
N

neha
Admin

[Log out](#)

Project

PROJECT



Projects

Create and manage every project in the platform.

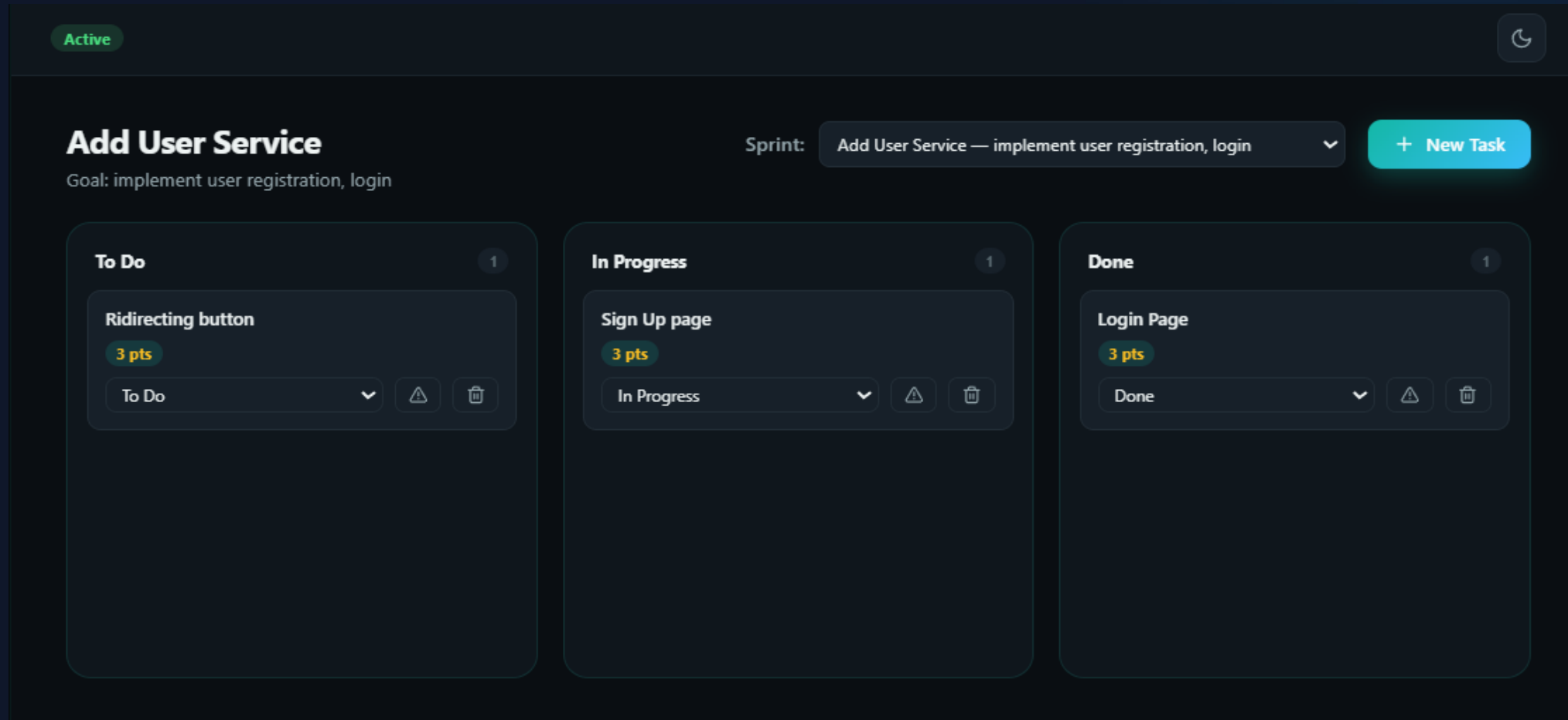
[+ New Project](#)

NAME	STATUS	TEAM	MANAGER	CREATED	
NeuroForge	Active	Frontend	neha	2026-08-18	Open

Need to rename, reassign a team, change status, or delete a project? Open it, then go to its Settings tab.

Kanban Board

Interactive Agile Kanban Board: Streamlining sprint workflows with dynamic task creation, story point estimation, and real-time lifecycle tracking.



Kanban Board

New task

Title

Ridirecting button

Description

User should be able to redirect from login to register page easily and vice versa

Story points

3

Assignee

Unassigned

Create task

Backlog

Product Backlog Management: Seamlessly organize unscheduled tasks and assign them directly to active sprints for automated synchronization with the Kanban board.

Active

Backlog

Tasks not yet assigned to a sprint.

+ Cancel

Sign Up page

3

Add to backlog

Unscheduled tasks (1)

Target sprint

Add User Service

≡ Login Page

3 pts

Add to sprint

Sprints & Milestones

Sprints & Milestones Management: Grouping iterative sprint cycles under structured milestones to maintain a clear roadmap and track target deadlines effectively.

Active

Sprints & Milestones

Plan sprints and group them under milestones for NeuroForge.

Sprints

+ New sprint

Add User Service

Viewing

implement user registration, login

2026-07-19 – 2026-07-25

Project & User Management

Add role based authentication

Integrate keycloak for IAM

2026-07-26 – 2026-08-01

Project & User Management

Kanban Board

Add Kanban Board to track the sprint status

2026-08-02 – 2026-08-08

Sprint & Task Management

Story Points

add tasks with story points

2026-08-10 – 2026-08-19

Sprint & Task Management

Milestones

+ New milestone

Project & User Management

Due 2026-07-19

Sprint & Task Management

Due 2026-08-01

Blockers

Blockers Management: Flagging critical bottlenecks to halt workflow operations and route issues to a dedicated resolution queue until cleared.

Active

Blockers

Flagged work in Add User Service

Add User Service

Open blockers (1)

⚠️ Rirecting button

not required now. · raised 8/24/2026

Resolve

Flag "Rirecting button" as blocked

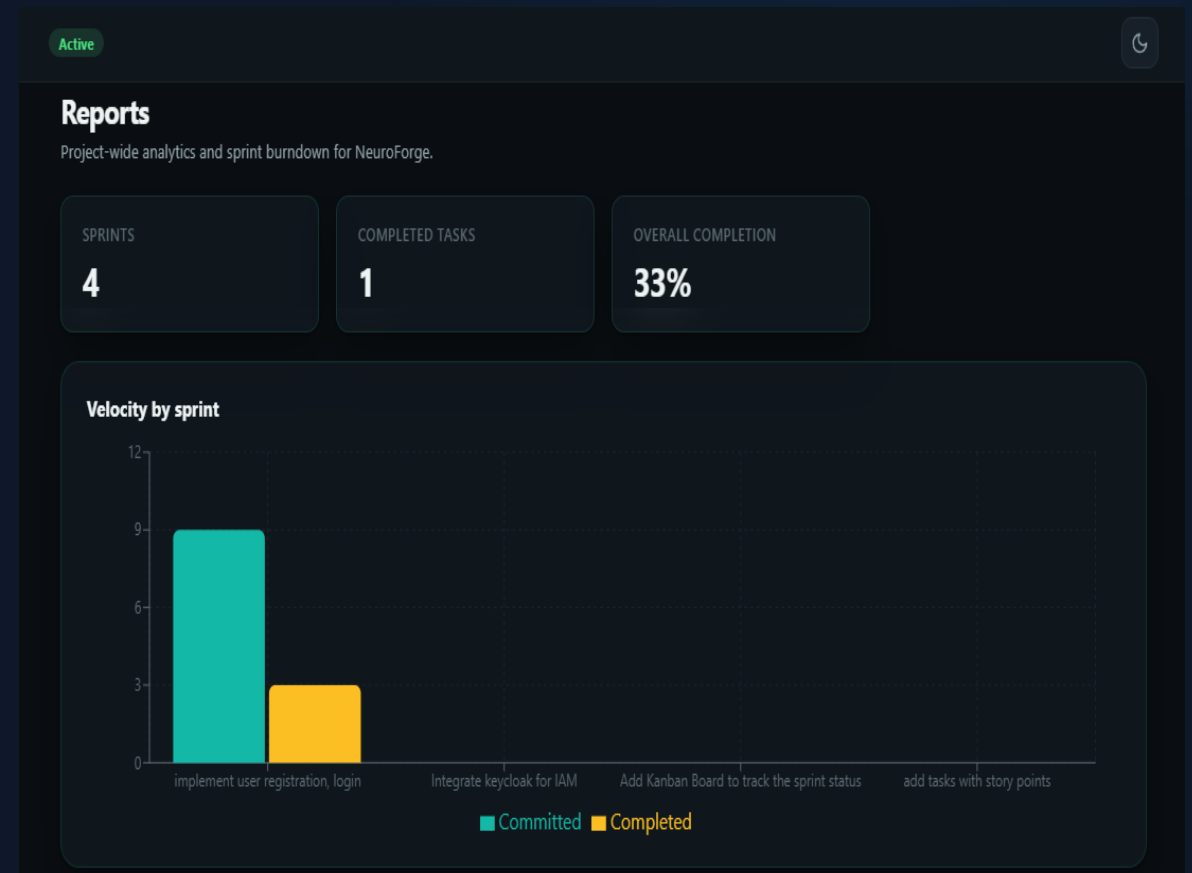
Why is it blocked?

not required now

Flag as blocked

Reports

Sprint Analytics & Burndown: Tracking velocity, committed vs. completed story points, and daily burn-down progress to monitor project health in real-time.



Settings

GitHub Integration & Project Settings: Linking remote repositories with secure webhooks and personal access tokens to power automated CI/CD workflows and manage granular project configurations.

Active

Project Settings

Reassign team, change status, or delete NeuroForge.

GitHub Repository

Repo owner

Neha-hub2345

Repo name

NeuroForge

Branch

main

Workflow file

ci-cd.yml

Personal access token (leave blank to keep current token)

.....

Update connection

Webhook secret

YXJV1a9vGwLIRMgx22Cj1f1nH8pePj-blJkCgC2jDmw

Add this as the WEBHOOK_SECRET Actions secret in your repo, alongside your existing CONTROLLER_URL, so NeuroForge can verify builds are really coming from your workflow.

Project name

NeuroForge

Status

ACTIVE

Team

Frontend

Manager

neha

No changes to save

DANGER ZONE

Delete this project

Removes the project and its association with sprints/milestones. This cannot be undone.

Delete project

Overview

Status

ACTIVE

Team

Frontend

Manager

neha

Project ID

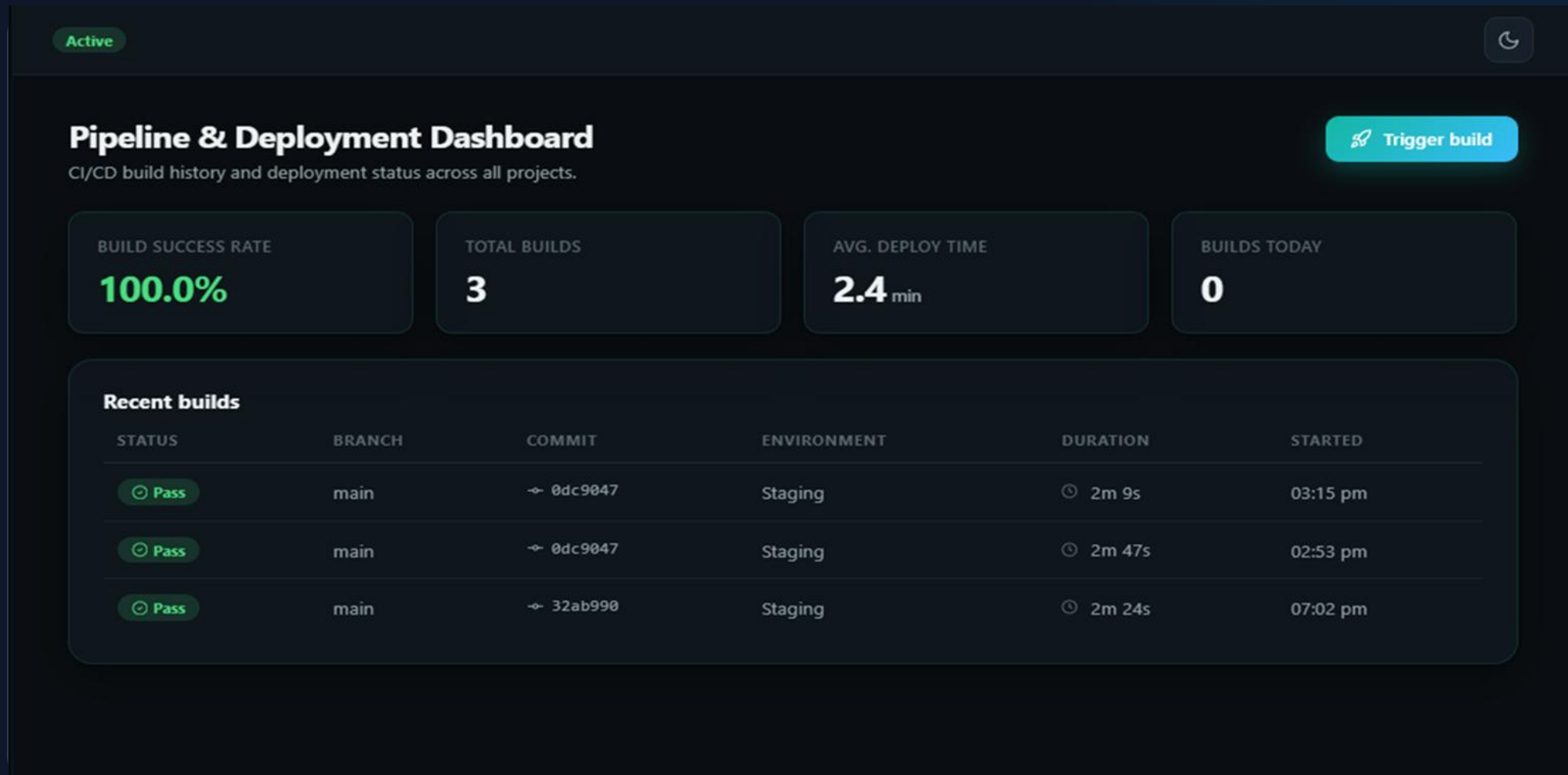
1

Created

2026-08-18

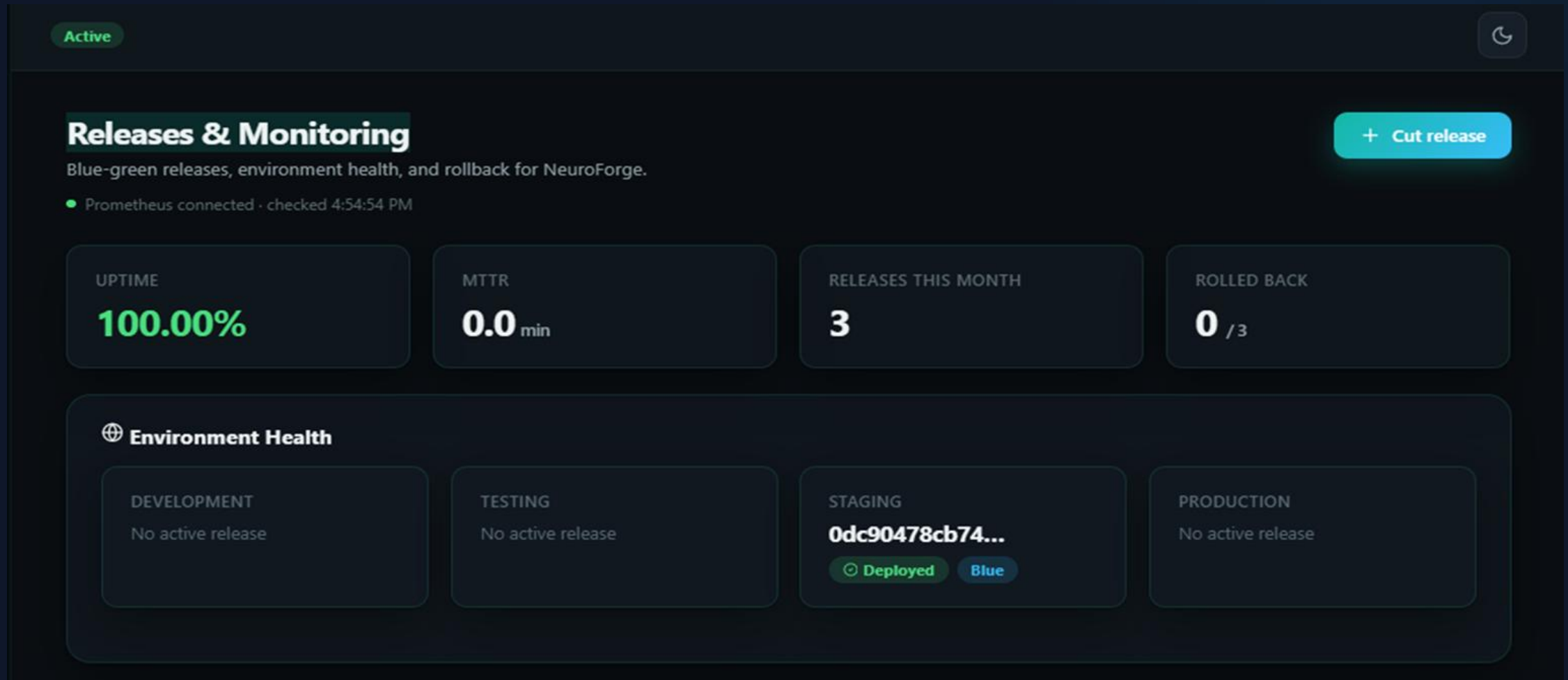
Pipelines & Deployments

CI/CD Pipeline & Deployment Dashboard: Automatically trigger builds and monitor real-time build histories, commit branches, and deployment metrics across environments.



Release & Monitoring

Release History & Blue-Green Deployments: Managing environment slots, tracking release versions, and maintaining zero-downtime deployment visibility.



Release & Monitoring

Alert Rules

+ New rule

No alert rules yet

Add one above to start testing alerts.

Active Alerts (0)

No active alerts

All monitored metrics are within thresholds.

Recent Logs



[Open full trail in Kibana](#)

```
04:55 pm INFO GET /api/kpi-history -> 200 (33ms)
04:55 pm INFO GET /api/alerts/rules -> 200 (8ms)
04:55 pm INFO GET /api/releases/active/STAGING -> 200 (30ms)
04:55 pm INFO GET /api/alerts -> 200 (12ms)
04:55 pm INFO GET /api/releases/active/PRODUCTION -> 500 (59ms)
04:55 pm INFO GET /api/releases -> 200 (63ms)
04:55 pm INFO GET /api/releases/kpi -> 200 (16ms)
04:55 pm INFO GET /api/releases/active/TESTING -> 500 (27ms)
04:55 pm INFO GET /api/observability/live -> 200 (37ms)
04:55 pm INFO GET /api/releases/active/DEVELOPMENT -> 500 (39ms)
04:55 pm INFO GET /api/observability/logs -> 200 (25ms)
04:55 pm INFO GET /api/releases/active/STAGING -> 200 (54ms)
```

Uptime & Pipeline Success — last 24h



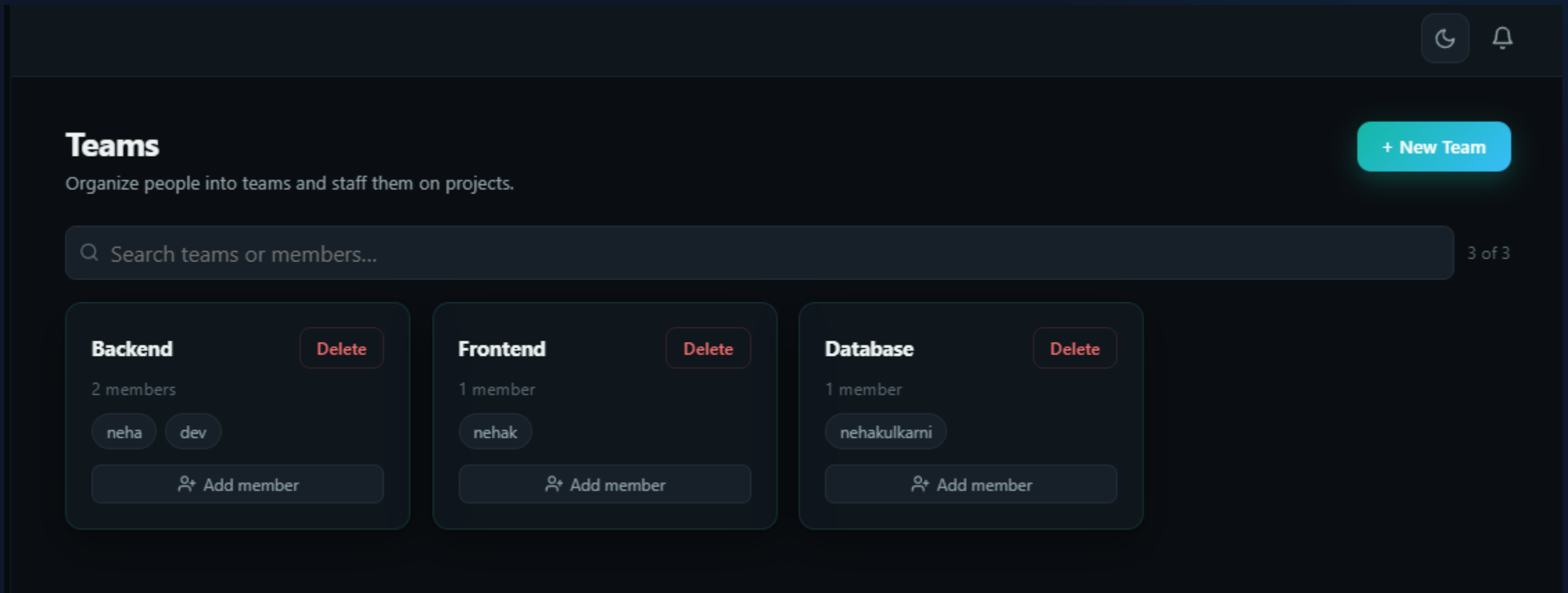
↔ Uptime % ↔ Pipeline Success %

Release history

VERSION	ENVIRONMENT	STATUS	SLOT	LIVE	RELEASED
0dc90478cb7478def9efeb95d06b10c7067fee4f	Staging	Deployed	Blue	Live	06:20 pm
0dc90478cb7478def9efeb95d06b10c7067fee4f	Staging	Superseded	Green	—	02:59 pm
32ab990f7832e13423759d1985069ae99cec1a76	Staging	Superseded	Blue	—	07:07 pm

Teams

Team Management & Staffing: Organizing registered platform users into specialized teams like Backend, Frontend, and Database to streamline project assignments.



Teams

Create a new team

Team name

Create team

Add a member to "Backend"

User

Select a user

Select a user

tester

devops

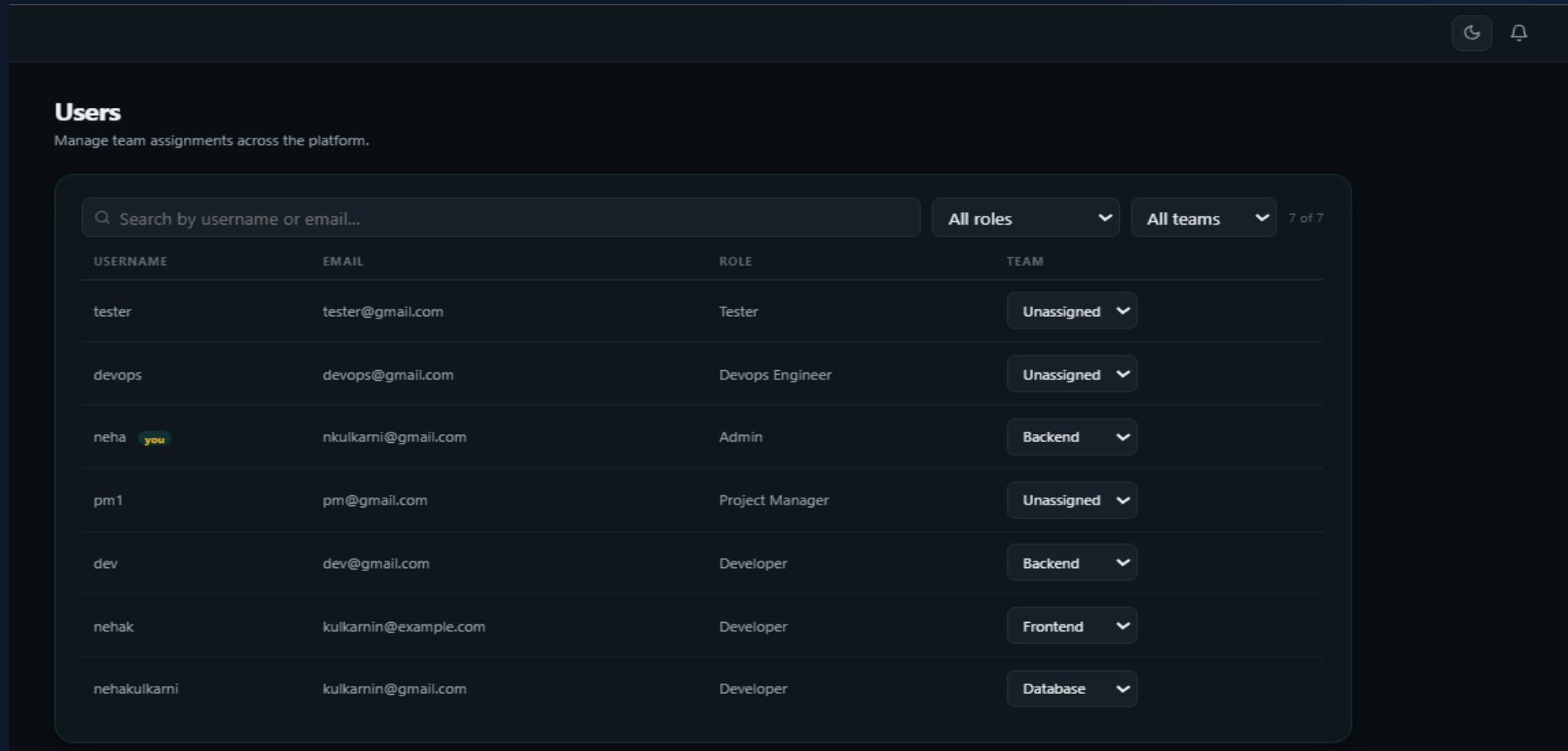
pm1

nehak (currently: Frontend)

nehakulkarni (currently: Database)

Users

User Management & Role Assignment: Centralized directory allowing administrators and project managers to view registered accounts, filter by roles, and manage team allocations dynamically.



USERNAME	EMAIL	ROLE	TEAM
tester	tester@gmail.com	Tester	Unassigned
devops	devops@gmail.com	Devops Engineer	Unassigned
neha you	nkulkarni@gmail.com	Admin	Backend
pm1	pm@gmail.com	Project Manager	Unassigned
dev	dev@gmail.com	Developer	Backend
nehak	kulkarnin@example.com	Developer	Frontend
nehakulkarni	kulkarnin@gmail.com	Developer	Database

AI Assistant

Groq AI Workspace Assistant: Integrated intelligence providing instant project summaries, pipeline health checks, and real-time environment insights.

The screenshot displays the 'Workspace Assistant' interface. At the top left is the title 'Workspace Assistant' with a star icon. At the top right is a 'Quick summary' button with a star icon. Below the title, a user message in a dark bubble says 'Give me a quick summary of everything important right now.' with a user icon. The assistant's response is in a light bubble, titled 'Current snapshot - key points' with a list icon. It contains four bullet points: 'Active project: NeuroForge (Frontend team, manager Neha) - the only active project.', 'Open blockers: 1 blocking issue (details not listed).', 'Pipeline health: 3 total builds, 100 % success, average deploy time 2.4 min; no builds have run today. Latest builds #3 and #2 both succeeded on `main` → STAGING.', and 'Release & environment health: Uptime 100 %, MTTR 0 min; 3 releases this month, none rolled back. Current STAGING deployment is commit `0dc90478cb7478def9efeb95d06b10c7067fee4f` on the **BLUE** slot (active). The previous release of the same commit is marked **SUPERSEDED**.' At the bottom, there is a text input field with the placeholder 'e.g. Which environments are unhealthy right now?' and a send button with a paper plane icon.

Workspace Assistant

Quick summary

Give me a quick summary of everything important right now.

Current snapshot – key points

- **Active project:** NeuroForge (Frontend team, manager Neha) – the only active project.
- **Open blockers:** 1 blocking issue (details not listed).
- **Pipeline health:** 3 total builds, 100 % success, average deploy time 2.4 min; no builds have run today. Latest builds #3 and #2 both succeeded on `main` → STAGING.
- **Release & environment health:** Uptime 100 %, MTTR 0 min; 3 releases this month, none rolled back. Current STAGING deployment is commit `0dc90478cb7478def9efeb95d06b10c7067fee4f` on the **BLUE** slot (active). The previous release of the same commit is marked **SUPERSEDED**.

e.g. Which environments are unhealthy right now?

Challenges Faced

Milestone1: Keycloak Security

Issue: Syncing JWT sessions securely with the Spring Boot backend.

Fix: Mapped Keycloak claims directly to PostgreSQL for real-time RBAC validation.

Milestone2:React Frontend State

Issue: UI lag during Kanban board drag-and-drop operations.

Fix: Optimized React state management to prevent excessive re-rendering.

Milestone3:GitHub Actions CI/CD

Issue: Securely triggering webhook from a local development setup.

Fix: Implemented secure ngrok tunneling and strict webhook secret verification.

Milestone4:Prometheus and Grafana

Issue: Monitoring tools consuming too much system RAM.

Fix: Capped Docker resource limits and optimized metric scraping intervals.

Team Members

Neha Kulkarni

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Sanika Mane

Shalini S



Thank You